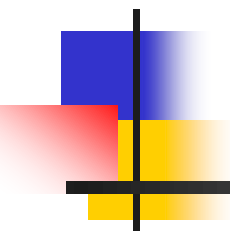


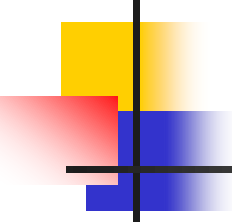
# **DATA STRUCTURES AND ALGORITHMS**



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## **Lecture Notes 11**

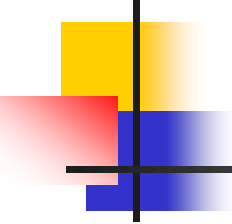
**Prepared by Pradit Pitaksathienkul**



# ROAD MAP

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- **GRAPH ALGORITHMS**
  - Definition
  - Representation of Graphs
  - Topological Sort
  - **Shortest Path Algorithms**
    - **Unweighted Shortest Paths**



# Shortest Path Algorithms

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- **Single Shortest Path Problem :**

*Given as input a weighted graph,  $G=(V,E)$ , and a distinguished vertex,  $s$ , find the shortest weighted path **from  $s$  to every other vertex in  $G$***

- Input is a weighted graph : associated with each edge  $(v_i, v_j)$  is a cost  $c_{i,j}$  to traverse the edge.
- Cost of a path  $v_1 v_2 \dots v_N$  (*weighted path length*) is

$$\sum_{i=1}^{N-1} c_{i,i+1}$$

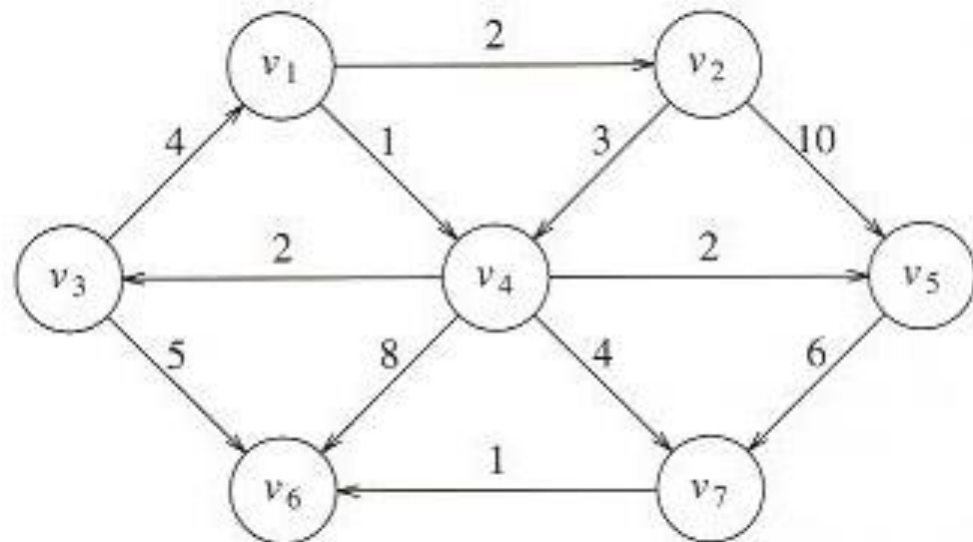
- Unweighted path length is merely the number of edges on the path,  $N-1$

# Shortest Path Algorithms

## Example :

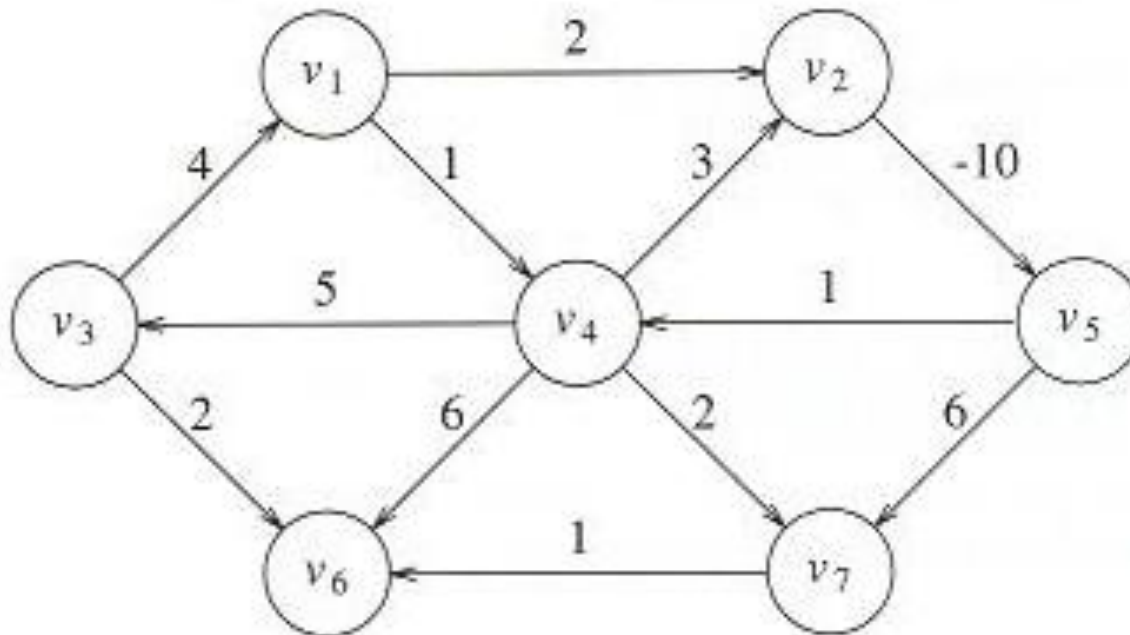
On the figure

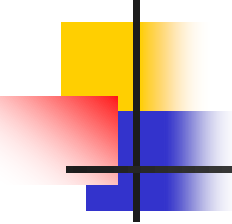
- Cost of shortest **weighted path** from  $v_1$  to  $v_6$  is 6
  - $v_1$  to  $v_4$  to  $v_7$  to  $v_6$
- Shortest **unweighted path** between these vertices is 2
  - $v_1$  to  $v_4$  to  $v_6$



# Shortest Path Algorithms

- Negative edges can cause problems!
- Path from  $v_5$  to  $v_4$  has cost 1 but a shorter path exists by following the loop  $v_5, v_4, v_2, v_5, v_4$ , has a cost -5
- Shortest path is undefined if there is a negative cost cycle





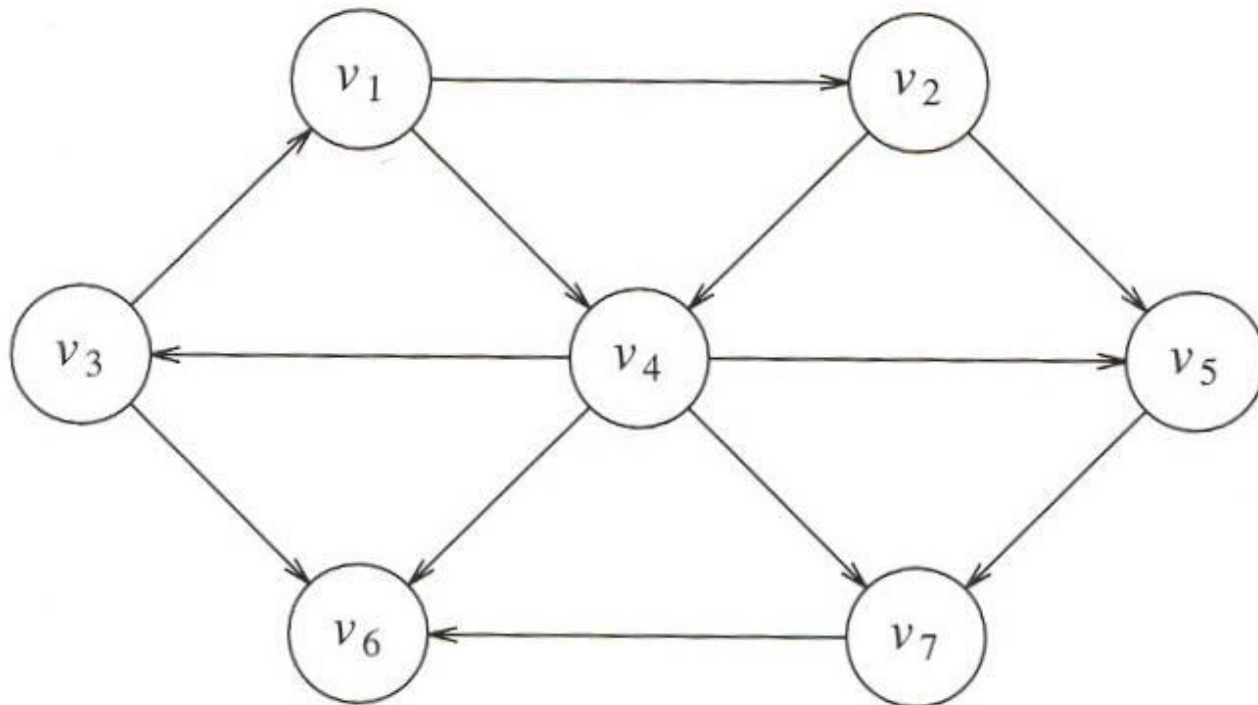
# Shortest Path Algorithms

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- **Shortest Path Algorithms**
  - **Unweighted Shortest Paths**

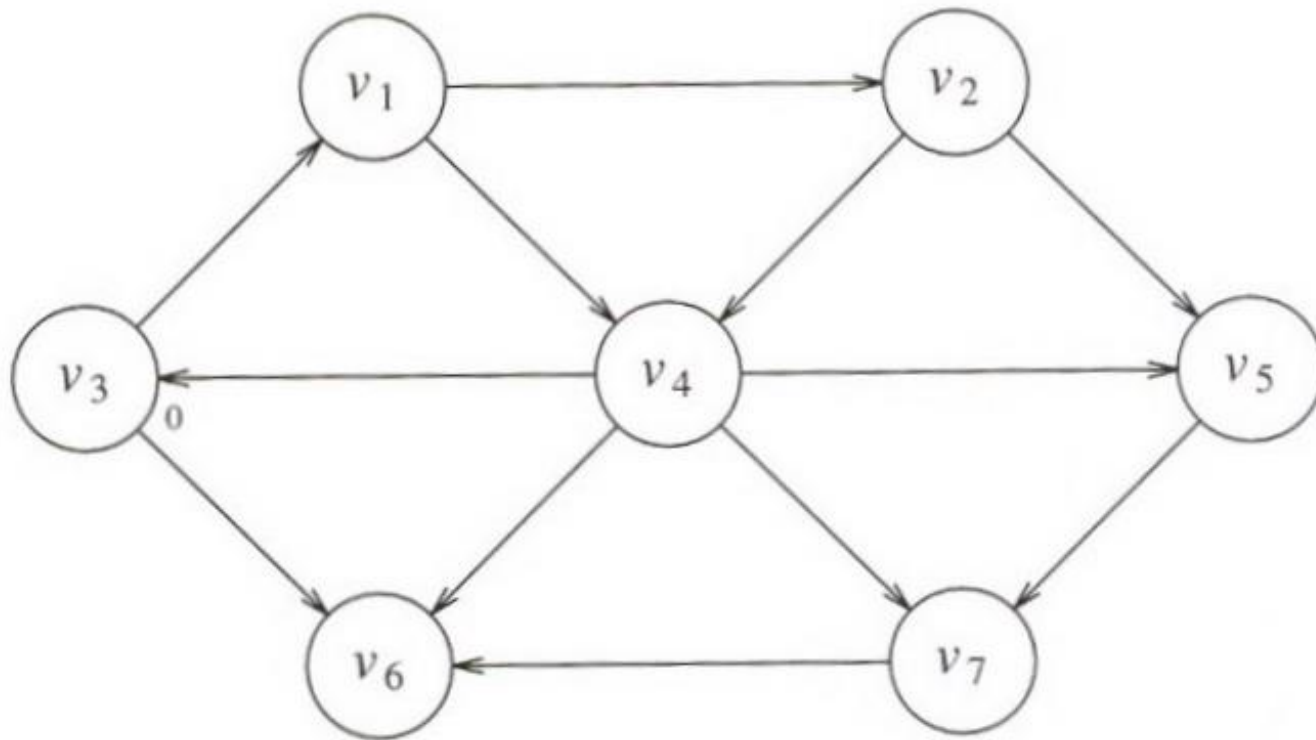
# Unweighted Shortest Paths

- We are only **interested in the number of edges** contained on the path
- There are **no weights** on the edges



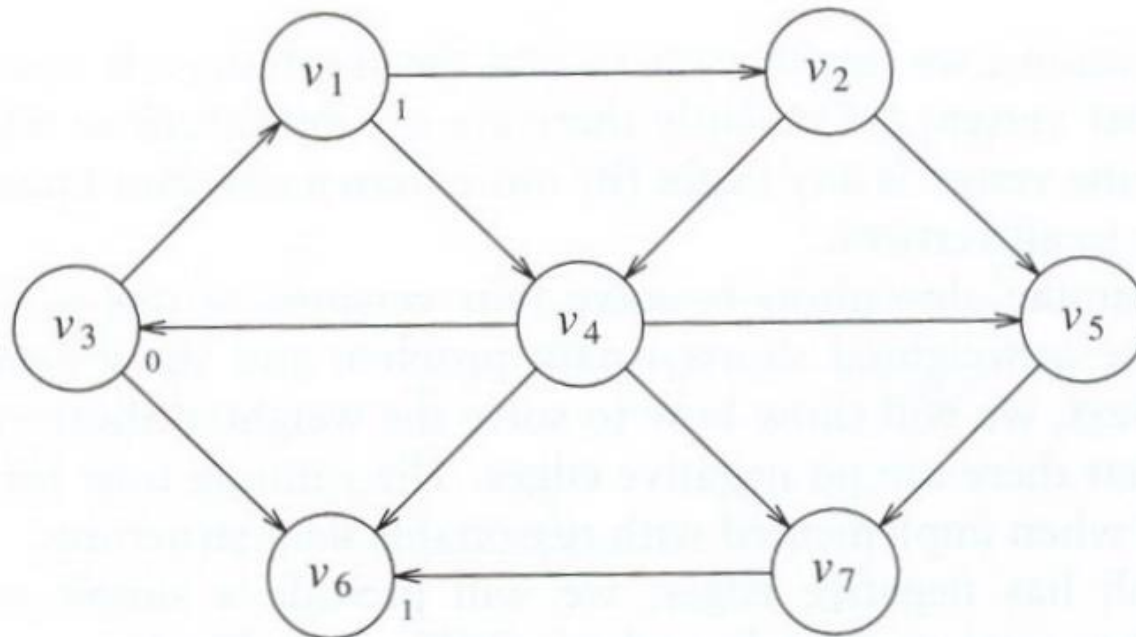
# Unweighted Shortest Paths

- Suppose **we choose  $s$  to be  $v_3$**
- Shortest path from  $s$  to  $v_3$  is then a path of length 0



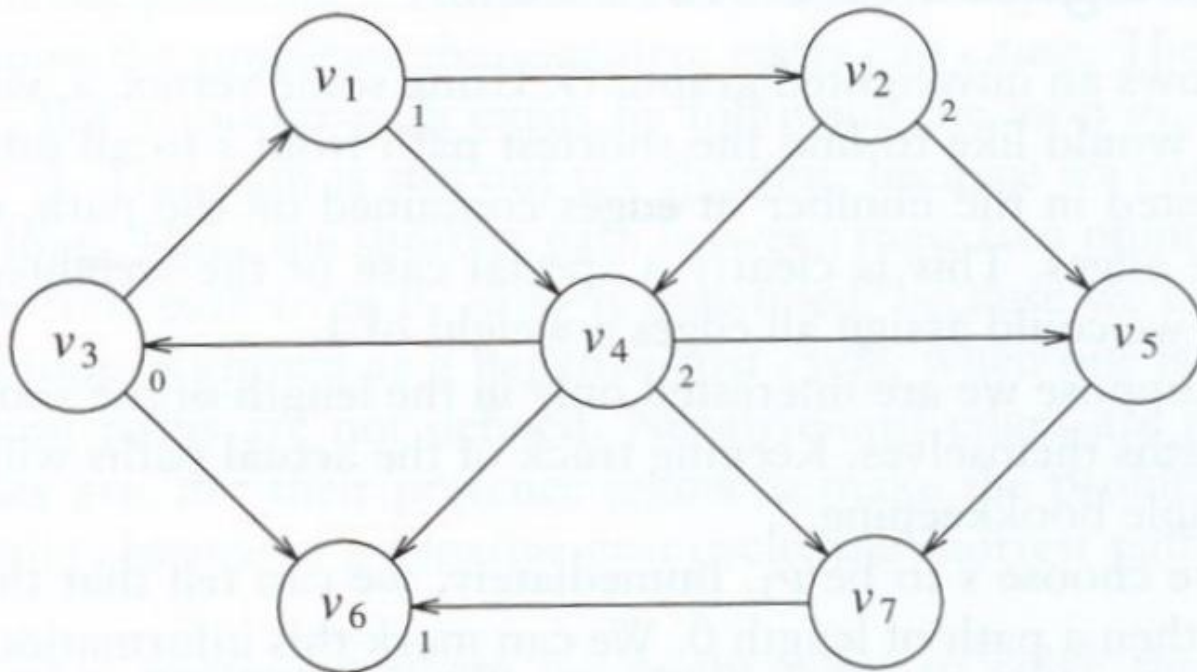
# Unweighted Shortest Paths

- We can start looking for all vertices that are a distance 1 away from  $s$ 
  - Can be found by looking at the vertices adjacent to  $s$
  - $v_1$  and  $v_6$  are one edge distance from  $s$



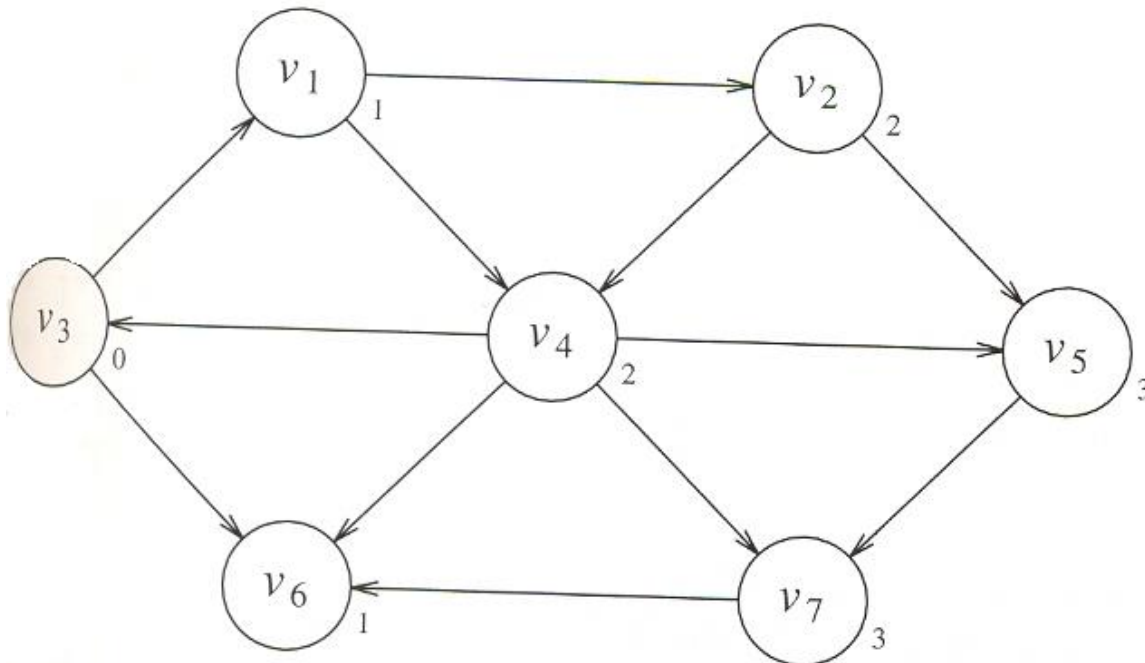
# Unweighted Shortest Paths

- We can find vertices whose shortest path from  $s$  is exactly 2
  - Look at all the vertices adjacent to  $v_1$  and  $v_6$  which are one edge from  $s$
  - So shortest path to  $v_2$  and  $v_4$  has length 2



# Unweighted Shortest Paths

- Finally we can find, by examining vertices adjacent to the recently evaluated  $v_2$  and  $v_4$ , that  $v_5$  and  $v_7$  have a shortest path of three edges.
- Now all vertices have been calculated



# Unweighted Shortest Paths

- Initial configuration of table used in unweighted shortest-path computation
- $D_v$  ย่อมาจาก คำว่า **distance** แปลว่า ระยะทาง ค่ามากไว้ก่อน 999
- $P_v$  ย่อมาจาก คำว่า **path** แปลว่า เส้นทาง 0

| $v$   | $known$ | $d_v$    | $p_v$ |
|-------|---------|----------|-------|
| $v_1$ | $F$     | $\infty$ | 0     |
| $v_2$ | $F$     | $\infty$ | 0     |
| $v_3$ | $F$     | 0        | 0     |
| $v_4$ | $F$     | $\infty$ | 0     |
| $v_5$ | $F$     | $\infty$ | 0     |
| $v_6$ | $F$     | $\infty$ | 0     |
| $v_7$ | $F$     | $\infty$ | 0     |

```

/*Pseudocode for unweighted shortest-path
  algorithm USE this Code*/
class Graph():
    def unweighted(s):
        q = Queue(NUM_VERTICES)
        define v, w as vertex
        q.enqueue( s )
        s.dist = 0

        while (!q.isEmpty()):
            v = q.items[0]
            q.dequeue()
            v.known = true

            for each w adjacent to v:
                if w.dist == INFINITY:
                    w.dist = v.dist + 1
                    w.path = v
                    q.enqueue (w)

```

| $v$   | Initial State |          |       | $v_3$ Dequeued |          |       | $v_1$ Dequeued  |          |       | $v_6$ Dequeued |          |       |
|-------|---------------|----------|-------|----------------|----------|-------|-----------------|----------|-------|----------------|----------|-------|
|       | $known$       | $d_v$    | $p_v$ | $known$        | $d_v$    | $p_v$ | $known$         | $d_v$    | $p_v$ | $known$        | $d_v$    | $p_v$ |
| $v_1$ | F             | $\infty$ | 0     | F              | 1        | $v_3$ | T               | 1        | $v_3$ | T              | 1        | $v_3$ |
| $v_2$ | F             | $\infty$ | 0     | F              | $\infty$ | 0     | F               | 2        | $v_1$ | F              | 2        | $v_1$ |
| $v_3$ | F             | 0        | 0     | T              | 0        | 0     | T               | 0        | 0     | T              | 0        | 0     |
| $v_4$ | F             | $\infty$ | 0     | F              | $\infty$ | 0     | F               | 2        | $v_1$ | F              | 2        | $v_1$ |
| $v_5$ | F             | $\infty$ | 0     | F              | $\infty$ | 0     | F               | $\infty$ | 0     | F              | $\infty$ | 0     |
| $v_6$ | F             | $\infty$ | 0     | F              | 1        | $v_3$ | F               | 1        | $v_3$ | T              | 1        | $v_3$ |
| $v_7$ | F             | $\infty$ | 0     | F              | $\infty$ | 0     | F               | $\infty$ | 0     | F              | $\infty$ | 0     |
| Q :   | $v_3$         |          |       | $v_1, v_6$     |          |       | $v_6, v_2, v_4$ |          |       | $v_2, v_4$     |          |       |

| $v$   | $v_2$ Dequeued |          |       | $v_4$ Dequeued |       |       | $v_5$ Dequeued |       |       | $v_7$ Dequeued |       |       |
|-------|----------------|----------|-------|----------------|-------|-------|----------------|-------|-------|----------------|-------|-------|
|       | $known$        | $d_v$    | $p_v$ | $known$        | $d_v$ | $p_v$ | $known$        | $d_v$ | $p_v$ | $known$        | $d_v$ | $p_v$ |
| $v_1$ | T              | 1        | $v_3$ | T              | 1     | $v_3$ | T              | 1     | $v_3$ | T              | 1     | $v_3$ |
| $v_2$ | T              | 2        | $v_1$ | T              | 2     | $v_1$ | T              | 2     | $v_1$ | T              | 2     | $v_1$ |
| $v_3$ | T              | 0        | 0     | T              | 0     | 0     | T              | 0     | 0     | T              | 0     | 0     |
| $v_4$ | F              | 2        | $v_1$ | T              | 2     | $v_1$ | T              | 2     | $v_1$ | T              | 2     | $v_1$ |
| $v_5$ | F              | 3        | $v_2$ | F              | 3     | $v_2$ | T              | 3     | $v_2$ | T              | 3     | $v_2$ |
| $v_6$ | T              | 1        | $v_3$ | T              | 1     | $v_3$ | T              | 1     | $v_3$ | T              | 1     | $v_3$ |
| $v_7$ | F              | $\infty$ | 0     | F              | 3     | $v_4$ | F              | 3     | $v_4$ | T              | 3     | $v_4$ |
| Q :   | $v_4, v_5$     |          |       | $v_5, v_7$     |       |       | $v_7$          |       |       | empty          |       |       |

How the data change during the unweighted shortest-path algorithm